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ACTIVE TENSIONING FOR ELECTROLYZER STACKS

Abstract

A method for sealing an electrolyzer cell may include applying a sealant between two layers of an electrolyzer cell and compressing the two layers towards each other. The method may further include flowing fluid through a flow field in the electrolyzer cell. The method may further include controlling a temperature of the fluid flowing through the flow field and controlling a pressure applied to the sealant by the compressing the two layers towards each other. The method may further include conforming the sealant to the two layers.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure generally relates to the manufacturing of seals for electrolyzers and fuel cells. More specifically, the present disclosure relates to methods and systems of sealing assemblies during the manufacture of electrolyzer stacks.

BACKGROUND

[0002] The drive for renewable energy solutions has resulted in substantial investments into water electrolysis or electrolyzer technologies. It is estimated that the electrolyzer market could increase to 300 GW over the next few decades, and power-to-gas is poised to become a multi-billion-dollar market for on-site electrolyzer systems. Electrolyzers use DC electricity to split water into hydrogen and oxygen. As one example, a proton electron membrane (PEM) electrolysis cell is a device which produces hydrogen and oxygen gas using DC electricity to electrochemically split water. A PEM cell contains an active area in which the presence of catalyst permits the reactions to take place. In the electrolysis cell, water enters the anode and is split into protons, electrons, and oxygen gas. The protons are conducted through the membrane, while the electrons pass through the electrical circuit. At the cathode, the protons and electrons recombine to form hydrogen gas.

[0003] Electrolyzers run at elevated temperatures and pressures and therefore must be sealed from the ambient environment. These operating conditions necessitate intensive processes to create functional electrolyzer cells, limiting the possible methods of manufacturing such electrolyzer cells.

[0004] Traditionally, electrolyzer stacks include a plurality of cells having plates or separators, structural frames and a sealant. The stacks, and/or the cell portions thereof, may be compressed together and heated to an optimal temperature, such as in an oven. The stacks or cells are then cooled down. However, once the stack or cells cools, the sealants may have insufficient pressure to maintain a seal between the adjacent portions (e.g., plates, separators and/or adjacent structural frames), especially at elevated temperatures. The stack and/or cells must then be resealed through recompressing and reheating. By repeating the heating and cooling cycle many times, the sealant eventually fully conforms to the electrolyzer stack, or the cell portions thereof, such that the seal may be maintained at operating temperatures.

[0005] The traditional process of cycling the separators and sealant(s) between high and low temperatures is slow, often requiring multiple hours per cycle over the course of multiple days. The traditional method is also labor intensive, requiring time and manpower to manually recompress the separators after each cycle. Thus, there is a need for improved methods and systems for manufacturing separator plates which have reduced time and labor investments when compared to traditional manufacturing processes.

SUMMARY OF THE INVENTION

[0006] A method for sealing an electrolyzer cell may include applying a sealant between two layers of an electrolyzer cell and compressing the two layers towards each other. The method may further include flowing fluid through a flow field in the electrolyzer cell. The method may further include controlling a temperature of the fluid flowing through the flow field and controlling a pressure applied to the sealant by the compressing the two layers towards each other. The method may further include conforming the sealant to the two layers.

[0007] A system for sealing an electrolyzer cell may include a fluid pump releasably connectable to a flow field of an electrolyzer cell. The electrolyzer cell may include two layers having a sealant therebetween, and the fluid pump may be configured to flow fluid through the flow field. The system may further include a thermal controller configured to control a temperature of the fluid flowed through the flow field and a tensioning device releasably connectable to the electrolyzer cell. The tensioning device may be configured to compress the two layers toward each other while the fluid flows through the flow field and the temperature of the fluid is controlled to conform the sealant to the two layers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the disclosure and together with the detailed description herein, serve to explain the principles of the disclosure. The drawings are thus only for purposes of illustrating preferred embodiments and are not to be construed as limiting the disclosure.

[0009] FIG. 1 depicts an exploded view of an internal subassembly of an electrolyzer stack in accordance with an aspect of the present disclosure;

[0010] FIG. 2 depicts a schematic view of the internal subassembly of FIG. 1 in accordance with an aspect of the present disclosure;

[0011] FIG. 3 depicts a schematic view of a system of the present disclosure and an electrolyzer cell of an electrolyzer stack in accordance with an aspect of the present disclosure;

[0012] FIG. 4 depicts a cross-sectional view of a portion of the system of FIG. 3 showing the internal subassembly of FIG. 1;

[0013] FIG. 5 depicts a schematic view of the system of FIG. 3 during performance of a voltage test, in accordance with an aspect of the present disclosure;

[0014] FIG. 6 depicts a schematic view of an electrolyzer stack including multiple instances of the electrolyzer cell of FIG. 3, in accordance with an aspect of the present disclosure;

[0015] FIG. 7 depicts a flow chart of a method of the present disclosure for use in operating the system of FIG. 3, in accordance with an aspect of the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0016] The present disclosure will be discussed hereinafter in detail in terms of various exemplary embodiments according to the present disclosure with reference to the accompanying drawings. In following the detailed description, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It will be obvious, however, to those skilled in the art that the present disclosure may be practiced without these specific details. In other instances, well-known structures are not shown in detail in order to avoid unnecessarily obscuring the present disclosure.

[0017] Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary, or the following detailed description. It is also understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

[0018] Referring to the drawings, wherein like reference numerals are used to indicate like or analogous components throughout the several views, and with particular reference to FIGS. 1-7, systems and methods and devices for manufacturing and sealing plates or separators of an electrolyzer stack are shown and discussed. The systems and methods and devices disclosed herein may include manufacturing, sealing and/or testing bipolar plates or separators prior to such plates being placed in operation in an electrolyzer stack.

[0019] FIG. 1 depicts various layers of an internal subassembly **100** of an electrolyzer cell of an electrolyzer stack including a cathode **105**, a membrane electrode assembly (MEA) **130**, and an anode **132**. The cathode **105** may include a cathode flow plate **110** at a first outer end **115** of the subassembly **100** and a cathode flow plate seal **120** on the inner side thereof. The anode **132** may include an anode flow plate **160** at a second outer end **165** of the subassembly **100** and an anode flow plate seal **150** on the inner side thereof. The MEA **130** may be located between the cathode flow plate seal **120** of the cathode **105** and the anode flow plate seal **150** of the anode **132**. In

embodiments, there may further be a first structural frame **172** and/or flow field **175** located between the anode flow plate seal **150** and the MEA **130**. Also in some embodiments, there may be a second structural frame **173** and/or flow field **176** located between the cathode flow plate seal **120** and the MEA **130**. In embodiments, the first structural frame **172** may include and/or peripherally surround the first flow field **175**, and the second structural frame **173** may include and/or peripherally surround the second flow field **176**, as shown in FIG. 1. The detailed description below is written as if the structural frames **172**, **173** include the flow fields **175**, **176**, but the present disclosure may be useful regardless of whether the structural frames **172**, **173** are separate from or combined with the flow fields **175**, **176**, respectively.

[0020] In some embodiments, there may be additional layers on one or both sides of the internal subassembly **100**, such as non-repeating hardware. For example, the non-repeating hardware may include end plates **168**, **169** (FIG. 3) on one or both sides of the internal assembly **100**. The non-repeating hardware may also include fluid connection plates, load support and/or distribution hardware (e.g., plates and/or tie rods), electrical busbar connections and/or electrical insulation hardware. In some embodiments, the cathode flow plate seal **120**, the anode flow plate seal **150**, and/or another seal may be applied directly to the non-repeating hardware to seal the non-repeating hardware to the cathode flow plate **110** and/or the anode flow plate **160**.

[0021] The cathode flow plate **110**, the anode flow plate **160**, and/or the structural frames **172**, **173** and/or flow fields **175**, **176** may be formed of a plastic, metal or another like material which one of ordinary skill would appreciate as being sufficient to allow for electrolysis to take place within a completed electrolyzer cell and/or electrolyzer stack. For example, the cathode flow plate **110**, the anode flow plate **160**, the structural frames **172**, **173** and/or the flow fields **175**, **176** may be made of materials which are known to be capable of withstanding the operating conditions under which an electrolyzer cell may be expected to operate.

[0022] The MEA **130** may include various layers including a membrane **140** (e.g., a polymer electrolyte membrane (PEM)) between a cathode side catalyst layer **125** of the cathode **105** and an anode side catalyst layer **135** of the anode **132**. A cathode side gas diffusion layer (GDL) **122** of the cathode **105** may be located between the cathode side catalyst layer **125** and the cathode flow plate **110**. An anode side gas diffusion layer **145** of the anode **132** may be located between the anode side catalyst layer **135** and the anode flow plate **160**.

[0023] In embodiments, the cathode flow plate seal **120** may be received between the cathode flow plate **110** and the second structural frame **173**, and the anode flow plate seal **150** may be received between the anode flow plate **160** and the first flow plate seal **172**. In further embodiments, the cathode flow plate seal **120** and/or the anode flow plate seal **160** may be located between different layers of the internal subassembly **100** than those just described. In some embodiments, the cathode flow plate **110** and the second structural frame **173** may be a single component, in which case the cathode flow plate seal **120** may be located between the combined components (e.g., the combined cathode flow plate **110** and the second structural frame **173**) and another layer of the internal subassembly **100**. Similarly, in some embodiments, the anode flow plate **160** and the first structural frame **172** may be a single component, in which case the anode flow plate seal **150** may be located between the combined components (e.g., the anode flow plate **160** and the first structural frame **172**) and another layer of the internal subassembly **100**. In some embodiments, the cathode flow plate seal **120** and the anode flow plate seal **150** may be received in a channel of an inner side of the cathode flow plate **110** and the anode flow plate **160**, respectively. In some embodiments, the cathode flow plate seal **120** and/or the anode flow plate seal **150** may be made of a thin plastic cut to a 2D profile of the mating interfaces of the layers to which they are to be applied.

[0024] When assembled, the internal subassembly, the end plates **168**, **169** and other non-repeating hardware, if any, may form a completed electrolyzer stack. Multiple electrolyzer cells (e.g., multiple of the internal subassembly **100**) may be combined to form an electrolyzer stack (FIG. 6) to be used for electrolysis.

[0025] FIG. 2 depicts the internal subassembly **100** of the electrolyzer cell, including the cathode **105**, the anode **132**, and the membrane **140** (e.g., a PEM). The cathode **105** includes the cathode side catalyst layer **125** which may in some embodiments include a platinum catalyst. The anode **132** includes the anode side catalyst layer **135** which may in some embodiments include an iridium catalyst.

[0026] Fluid (e.g., water) may be flowed into the anode **132**, which is then electrochemically split into oxygen gas and hydrogen ions. The hydrogen ions cross the membrane **140** and recombine with electrons to form hydrogen gas.

[0027] Turning to FIGS. 3-6, a system **1** of the present disclosure will now be described. With reference to FIG. 3, the system **1** for manufacturing and/or sealing plates or separators of an electrolyzer stack may receive two plates or separators **170**, **171** (e.g., the cathode flow plate **110** and/or the anode flow plate **160**) to be sealed to two structural frames and/or flow fields **172**, **173** using a sealant **190** (e.g., the cathode flow plate seal **120** and/or the anode flow plate seal **150**). As depicted in FIG. 3, the MEA **130** may be located between structural frames **172**, **173** which may each include flow fields. In some embodiments, the flow fields **175**, **176** may include lattice-like structures within and/or along which fluid may flow and/or traverse.

[0028] The MEA **130** is depicted vertically between and adjacent structural frames **172**, **173** for convenience but portions of the MEA **130** may be received in interior open portions of structural frames **172**, **173**. For example, smaller outer portion(s) **178** of the MEA **130** may be received between and/or against outer portions of structural frames **172**, **173** while inner larger portion(s) **179** of the MEA **130** may be received in, between and/or against the interior open portions of structural frames **172**, **173**. In embodiments, the inner larger portion(s) **179** may be received such that they are adjacent to and/or abutting one or both of the flow fields **175**, **176**, which may also be received in the interior open portions of the structural frames **172**, **173** (FIG. 1).

[0029] The sealant **190** may be located between the two separators **170**, **171** and the two structural frames and/or flow fields **172**, **173** and may be conformed thereto to bind the two separators **170**, **171** to the two structural frames **172**, **173** and/or other components of the internal subassembly **100** together by using the systems and methods described herein. For example, the sealant **190** may be located between a first separator **170** of the two separators **170**, **171** and a first structural frame and/or flow field **172** of the two structural frames and/or flow fields **172**, **173** and may be conformed thereto to bind the first separator **170** to the first structural frame **171**. Similarly, the sealant **190** may (also) be located between a second separator **171** of the two separators **170**, **171** and a second structural frame and/or flow field **172** of the two structural frames and/or flow fields **172**, **173** and may be conformed thereto to bind the second separator **171** to the second structural frame **172**. In embodiments, there may also be a sealant (e.g., the sealant **190** or a different sealant) between layers of the MEA **130**. In further embodiments, the sealant **190** may be located between various other layers of the internal subassembly **100** described above, or between any such layer(s) and the non-repeating hardware (e.g., the end plates **168**, **169**), depending on the layout and/or operational requirements of a particular electrolyzer cell being manufactured. Components such as the cathode gas diffusion layer **122**, the cathode side catalyst layer **125**, the anode side catalyst layer **135**, the anode side diffusion layer **145**, among other layers and/or components, may be between the two separators **170**, **171** during operation of the system as described below, as shown in FIG. 4. Once a sealant (e.g., the sealant **190**) has been conformed through operation of the system as described below, the completed product may be an electrolyzer cell sufficient to be incorporated into an electrolyzer stack and to be used for electrolysis.

[0030] By “conformed,” it is meant that the sealant **190** molds to mating interfaces of the two separators **170**, **171** and the two structural frames **172**, **173** (e.g., to various inner surfaces (not shown) of the two separators **170**, **171** and/or the two structural frames **172**) to form a seal sufficient to isolate an electro-chemical portion of the cell (e.g., the internal subassembly **100**) from an exterior of the two separators **170**, **171**. The conforming of the sealant **190** may thus be

necessary for electrolysis to take place within a completed electrolyzer cell **195** of an electrolyzer stack **196** (FIGS. 5-6; discussed in more detail below). The conforming of the sealant **190** to the mating surfaces of the two separators **170**, **171** and/or the two structural frames **172**, **173** may thus be necessary for reliable operation of the electrolyzer cell **195** throughout its useful lifetime. If the sealant **190** were to not be properly conformed and/or conditioned, the quality of the seal would deteriorate over time which may lead to an early failure of the device. Additionally, such failure may present itself in a way that creates hazards to the environment in which the electrolyzer cell **195** may operate. A plurality of nuts and/or bolts **177** may connect the various elements of the electrolyzer cell **195** for example, on the end plates **168**, **169** and/or on opposite sides of the two separators **170**, **171**, as shown in FIG. 3.

[0031] Turning back to FIG. 3, the system **1** may include a compressor or tensioning device **200** that is removably connectable to the two separators **170**, **171**, the end plates **168**, **169** and/or one or multiple bolts of the plurality of bolts **177**. In some embodiments, the tensioning device **200** may be a hydraulic bolt tensioner, as shown in FIG. 3. However, the tensioning device **200** may be any device which one skilled in the art would know and understand to be sufficient to apply pressure to the two separators **170**, **171** and/or the two structural frames **172**, **173** to compress them towards each other and/or the sealant **190**. For example, force could be applied via fluid power (either hydraulics or pneumatics via a press or bolt tensioner), electrical means (electric motor, servo actuator), a displacement limit (where the hardware cannot move and force is generated through expansion forces, e.g., thermal heating or hydration expansion), or gravity (e.g., setting weights on top of the assembly).

[0032] In the example, the tensioning device **200** may apply pressure to the two separators **170**, **171**, the two structural frames **172**, **173** and/or the sealant **190** in at least one or both of a first direction A and/or a second direction B. The first direction A may be the direction from a first separator **170** of the two separators **170**, **171** towards a second separator **171** of the two separators **170**, **171** (e.g., perpendicularly to longitudinal dimensions of separators **170**, **171**). The second direction B may be a direction from the second separator **171** towards the first separator **170** (e.g., perpendicularly to longitudinal dimensions of separators **170**, **171**). The compression applied to the two separators **170**, **171** and/or the two structural frames **172**, **173** may be useful in controlling a creeping or conforming of the sealant **190** to any of the two separators **170**, **171** and/or the two structural frames **172**, **173** to bind the two separators **170**, **171** and the two structural frames **172**, **173** together.

[0033] In some embodiments, the tensioning device **200** may by itself apply sufficient pressure or compression to effectuate the conforming and/or binding of the sealant **190** to the two separators **170**, **171** and/or the two structural frames **172**, **173**. In further embodiments, such as where the tensioning device **200** is a hydraulic bolt tensioner as shown, the tensioning device **200** may be configured to apply pressure only to a first bolt **177a** of the plurality of bolts **177**, which may or may not be sufficient to carry out operation of the system as described below, depending on production requirements. In such an embodiment where the single tensioning device **200** is insufficient to conform the sealant **190**, a plurality (not shown) of the tensioning device **200** may be used to simultaneously tighten any or all of the plurality of bolts **177**, and/or to apply pressure to the two separators **170**, **171** and/or the two structural frames **172**, **173** to compress the components towards each other and/or to conform the sealant **190** therebetween.

[0034] In embodiments, the compressor or tensioning device **200** may include a pressure controller **202** and/or a pressure sensor **204**. The pressure controller **202** may be configured to control the tensioning device **200** to actively increase, decrease, or maintain a tensioning, compression and/or pressure applied to the two separators **170**, **171** and/or the two structural frames **172**, **173** (e.g., to the first bolt **177a**) by the tensioning device **200** during operation of the system **1**. The pressure sensor **204** may be configured to measure the tensioning, compression or pressure applied to the two separators **170**, **171** and/or the two structural frames **172**, **173** via the bolt tensioning device

200. In some embodiments, the pressure sensor **204** may be directly attached to or apart of the tensioning device **200**. In other embodiments, the pressure sensor **204** may be located on or between the two separators **170**, **171** and/or the two structural frames **172**, **173**. In some embodiments, the pressure tensioning device may further include a pressure release or control valve (not shown) to increase the safety of the system **1** during operation. The pressure control valve may be configured to limit the maximum force and/or pressure that could be exerted by the tensioning device **200**. The pressure control valve may be a single or multi-use device that is controlled by a spring, electronically, or by precision bursting of material (e.g., rupture disc).

[0035] In some embodiments, the compression and/or pressure applied to the two separators **170**, **171**, the two structural frames **172**, **173**, and/or the sealant **190** may be controlled manually, such as by an operator. In such an embodiment, the operator may receive information regarding the compression and/or pressure applied to the two separators **170**, **171**, the two structural frames **172**, **173**, and/or the sealant **190** by the pressure sensor **204**, and the operator may use that information to regulate the system **1** via the pressure controller **202**.

[0036] In other embodiments, the pressure controller **202** and/or the pressure sensor **204** may be configured to dynamically, and/or automatically control the tensioning, compression and/or pressure applied to the two separators **170**, **171**, the two structural frames **172**, **173**, and/or the sealant **190**. By dynamically and/or automatically, it is meant that as changes occur in the system **1** during operation, the system **1** adjusts without input from an operator to regulate and/or maintain optimal process conditions based on feedback received from sensors (e.g., the pressure sensor **204**). For example, during operation of the system **1**, the sealant **190** may creep or conform to the two separators **170**, **171** and/or the two structural frames **172**, **173** as the result of compression and/or heating (described in more detail below). As the sealant **190** conforms, the sealant **190** may change shape and/or thickness, which may alter the amount of pressure and/or compression applied to the two separators **170**, **171**, the two structural frames **172**, **173**, and/or the sealant **190**. As the sealant **190** conforms, the pressure sensor **204** may detect a change in the amount of pressure and/or compression being applied (e.g., to the two separators **170**, **171** and/or the two structural frames **172**, **173**). Upon detecting such a change, the pressure sensor **204** may be configured to send a signal to the pressure controller **202** directing the pressure controller **202** to modify the compression and/or pressure being applied in order to regulate and/or maintain compression within a prescribed range of pressure.

[0037] In some embodiments, the tensioning device **200** may also be controlled by sensors monitoring other characteristics of the stack such as measurement of a distance between the two separators **170**, **171**, between one or both of the two separators **170**, **171** and one or both of the two structural frames **172**, **173**, between one or both of the two separators **170**, **171** and one or both of the end plates **168**, **169** and/or between various other components mentioned above as the components are pressed towards each other and the distance between those components is measurably changed. In order to prevent damage to certain materials, the rate of compression may have to be controlled. While a force-based compression may still be desirable for optimal product quality, limiting the pressure to account for other phenomenon may be critical.

[0038] While the pressure controller **202** is shown in FIGS. **3** and **5** as being part of or attached to the tensioning device **200**, it is noted that the pressure controller **202** may be a remote device such as a PC, cell phone, or other electronic or analog (e.g., in the case of an operator performing the compression a standard pressure gauge may be used) device capable of sending and receiving signals and/or information remotely.

[0039] Consistent regulation of the compression applied to the two separators **170**, **171**, the two structural frames **172**, **173**, and/or the sealant **190**, either manually or automatically as just described, may be particularly useful in reducing process times during the manufacturing of electrolyzer stacks. Specifically, active control and/or regulation over the compression being applied may reduce the need to cool down the two separators **170**, **171**, the two structural frames

172, 173, and/or the sealant **190** during manufacture by maintaining the compression and/or load applied thereto within a proscribed range throughout the process. By reducing the need to cool materials during production (e.g., the two separators **170, 171**, the two structural frames **172, 173**, and/or the sealant **190**), the manufacturing times for producing electrolyzer stacks may be greatly reduced. In some cases, the lack of a need to cool components may trim hours or even days off of expected or typical processing times.

[0040] With continued reference to FIG. **3**, the system **1** may further include a fluid circulation device or fluid pump **250**. In embodiments, the fluid pump **250** may be removably connectable to one or both of the two structural frames **172, 173**. In the example shown, the fluid pump **250** may be removably connected to a fluid port **180** of the first structural frame **172**. The fluid pump **250** may be configured to pump and/or flow fluid (e.g., water) through one or multiple channels and/or the flow field(s) **175, 176** (not shown) of the two structural frames **172, 173** during operation of the system **1**. For example, such a channel or flow field may be a channel or flow field (e.g., the flow fields **175, 176**) for flowing water through the electrolyzer cell **195** (e.g., internal subassembly **100**, FIG. **1**) of the electrolyzer stack **196** to electrochemically split the water during operation.

[0041] The fluid pump **250** may include a pump controller **252**, a flow sensor **254** and/or a temperature or thermal sensor **256**. In embodiments, the pump controller **252** may actively control a flow (e.g., a rate of flow of water) of fluid into the fluid port **180** and/or through the one or multiple channels and/or the flow field(s) **175, 176** of the two structural frames **172, 173** during operation of the system **1**. Also in embodiments, the pump controller **252** may control a temperature of the fluid flowed through the one or multiple channels and/or the flow field(s) **175, 176**. For example, the pump controller **252** may be responsible for heating the fluid which, upon being flowed through the one or multiple channels and/or the flow field(s) **175, 176**, transfers heat to the two separators **170, 171**, the two structural frames **172, 173**, and/or the sealant **190** to facilitate the conforming of the sealant **190** to the two separators **170, 171** and/or the two structural frames **172, 173**.

[0042] In some embodiments, the fluid pump **250** may also include a filter or set of filters (not shown). During manufacturing (e.g., during operation of the system **1**), the filters may filter and/or remove debris and/or contaminants, preventing those materials from being flowed through the one or multiple channels and/or the flow field(s) **175, 176**. Following operation of the system **1**, the electrolyzer cell **195** may be used for electrolysis. During electrolysis, the filter or set of filters may help to maintain the electrical properties of the circulating fluid, such as by maintaining a high electrical resistance via removing materials which are electrically conductive from the fluid.

[0043] In some embodiments, the pump controller **252** may be operated manually, such as by the operator. In such an embodiment, the operator may be fed information regarding the flow rate of the fluid and/or the temperature of the fluid flowing through the one or multiple channels and/or the flow field(s) **175, 176** by the flow sensor **254** and/or the temperature sensor **256**, which the operator may use to regulate the system **1**. In some embodiments, the operator may regulate the system **1** via the pump controller **252**. In other embodiments, the operator **1** may regulate the system in the absence of the pump controller **252** by manually and directly adjusting components of the pump **250** during operation.

[0044] In other embodiments, the pump controller **252** may be configured to dynamically, and/or automatically control the flow rate and/or the temperature of the fluid flowed through the one or multiple channels and/or the flow field(s) **175, 176** based on feedback received from the flow sensor **254** and/or the temperature sensor **256** to regulate and/or maintain optimal process conditions. For example, if the temperature sensor **256** senses that the temperature of the fluid being flowed through the one or multiple channels and/or the flow field(s) **175, 176** is lower than optimal temperatures for manufacturing, the temperature sensor **256** may send a signal to the pump controller **252** directing the pump controller **252** to cause a heater **253** to increase the temperature of the fluid. In some embodiments, the pump **250** may act as the heater, or may have the heater

(e.g., heater **253**) incorporated therein, as shown in FIG. 3. In other embodiments, the heater may be its own, separate component of the system which heats the fluid before and/or as the fluid is flowed through the one or multiple channels and/or the flow field(s) **175, 176**.

[0045] The heater (e.g., heater **253**) must be compatible with and capable of withstanding all contaminants and process conditions (flow, temperature, etc.) present during manufacturing or operation of the system **1**, but the specific type of heater may be any heater which one skilled in the art would appreciate as being sufficient to heat the fluid as described (e.g., electrically powered, gas powered, etc.). The size of the heater may be of particular concern. For example, if the heater is too small the process may not be efficient enough to provide an advantage over other systems and methods of manufacturing electrolyzer cells. On the other hand, if the heater is too large the controllability of the system **1** to conform the sealant **190** may be poor. Where the heater is too large, applying too rapid of a temperature change (thermal shock) may cause damage to the various components of the electrolyzer cell **195** described above.

[0046] In embodiments, the pump controller **252** may be configured to gradually increase the temperature of the fluid flowed through the system to a process temperature which is within a proscribed range. The proscribed range may be the range of temperatures which best allows for the sealant **190** to creep and/or conform to the two separators **170, 171** and/or the two structural frames **172, 173**. The proscribed range of temperatures may be optimized based on design specifications and/or working conditions. The process and equipment need to be capable of efficiently functioning at or above a maximum temperature of the proscribed range of temperatures during operation of the system **1**. The material used as a sealant (e.g., the sealant **190**) may vary widely and may include rubber, Teflon, or plastic materials that are suited to the application and environment. It's important that the proscribed range of temperatures, and other operating conditions, be tuned properly with the fluid flowed through the one or multiple channels and/or the flow field(s) **175, 176** by the pump **250**, as phase changes (freezing/boiling) may expose the electrolyzer cell **195** and/or the various components thereof to conditions that were not designed for, which may cause damage. Sealing materials (e.g., materials used as and/or in the sealant **190**) are those that are amenable to the operating conditions of the completed electrolyzer stack **196**. The sealing materials (e.g., materials used as and/or in the sealant **190**) may therefore have properties such as excellent hydrolytic stability, elastic properties when under pressure, the ability to be easily processed (cut, dispensed, placed etc. as an electrolyzer stack is assembled). Perhaps most important is that the properties of such materials do not change over time throughout the constant temperature and pressure cycling that occurs in an electrolyzer stack (e.g., the electrolyzer stack **195**).

[0047] While the pump controller **252** is shown in FIGS. 3 and 5 as being part of or attached to the fluid pump **250**, it is noted that the pump controller **252** may be a remote device such as a PC, cell phone, or other electronic device capable of storing data, carrying out programmable instructions, and/or sending and receiving signals and/or information remotely.

[0048] The process of flowing heated fluid through the one or multiple channels and/or the flow field(s) **175, 176** may be aided by the compression applied to the two separators **170, 171**, the two structural frames **172, 173**, and the sealant **190** by the tensioning device **200**, as described above. As the sealant **190** conforms from the heat and/or compression applied to the two separators **170, 171**, the two structural frames **172, 173**, and/or the sealant **190**, the sealant **190** may change in shape and/or thickness, causing a change in the pressure being applied. The tensioning device **200**, via the pressure controller **202** and/or the pressure sensor **204**, may detect such a change and compensate, for example, by increasing the pressure until the pressure being applied is within a proscribed range of pressure(s).

[0049] The active regulation of the pressure, flow rate and/or thermal profile of the system **1** during manufacture of a subassembly (e.g., internal subassembly **100**) of an electrolyzer cell of an electrolyzer stack, as described above, may function to reduce process times to produce

electrolyzer stacks by reducing or eliminating the need to cool and reheat components of the system **1**. The active regulation thus permits a system (e.g., system **1**) for compressing and sealing electrolyzer separator plates to operate for longer periods of time at optimal process temperatures, without damaging any components of the electrolyzer stack being manufactured, which may result in a better, more consistent seal between such components (e.g., the two separators **170**, **171**, the two structural frames **172**, **173** and/or the sealant **190**). Additionally, the active regulation of the system **1** as just described may allow for the manufacturing process to be expedited, speeding up processes that occur naturally to be fast enough to enable serial production of electrolyzer cells and/or stacks (e.g., the electrolyzer cell **195** and/or the electrolyzer stack **196**).

[0050] Advantageously, the present disclosure permits for testing of electrolyzer cells (e.g., the electrolyzer cell **195**) and/or stacks (e.g., the electrolyzer stack **196**) during manufacture to ensure the electrolyzer stacks are in condition to be used (e.g., during electrolysis) and in compliance with regulatory requirements. For example, after the sealant **190** has conformed to the two separators **170**, **171** and/or the two structural frames **172**, **173**, such that an electrolyzer cell **195** of an electrolyzer stack has been formed from the two separators **170**, **171**, the two structural frames **172**, **173**, the sealant **190**, and/or the various other components of the internal subassembly **100** (FIG. 1), the electrolyzer cell **195** and/or the electrolyzer stack **196** may be tested to ensure the electrolyzer stack **196** is in operational condition.

[0051] In embodiments, one test which may be performed on the electrolyzer cell **195** and/or the electrolyzer stack **196** is a leak-check. The leak-check may be performed during manufacture at a variety of pressures and temperatures. In embodiments, the leak check may be performed by forcing fluid (e.g., water) or gas through the one or multiple channels and/or the flow field(s) **175**, **176** (not shown) of the electrolyzer cell **195** and/or the electrolyzer stack **196** to determine whether unwanted fluid communication may be present between any of the channels of the one or multiple channels (e.g., between portions of the flow field(s) **175**, **176**), or between any of the channels (e.g., between portions of the flow field(s) **175**, **176**) and a space exterior to the electrolyzer cell **195** and/or the electrolyzer stack **196**. In some embodiments, the fluid or gas may be forced through the one or multiple channels and/or the flow field(s) **175**, **176** using the fluid pump **250**. In other embodiments, a different leak-checking device may be used, the leak-checking device being any such device which one of ordinary skill in the art would appreciate as being able to detect leaking fluid or gas. The proscribed fluid or gas may be pressurized above normal atmosphere.

[0052] Additionally, the leak check may be performed across multiple different channels of the one or multiple channels (e.g., fluid paths and/or portions of the flow field(s) **175**, **176**), and in some embodiments the check between multiple channels may be performed concurrently. By design, the one or multiple channels and/or the flow field(s) **175**, **176** may be completely independent, or interconnected with media (e.g., the Membrane Electrode Assembly **140**) that will allow for connection. Detection of leaks can be done by a multitude of ways including bubbles per minute, drips per minute, fluid or gas displaced in a given amount of time, mass flow meter, differential pressure, pressure decay, or any other metric which one skilled in the art would know and understand to provide feedback on whether a leak is present. The fluid or gas may be flowed through the one or multiple channels and/or the flow field(s) **175**, **176** to check for a leak at various pressures and/or flow rates.

[0053] In embodiments, another test which may be performed on the electrolyzer cell **195** and/or the electrolyzer stack **196** is a voltage test, as shown in FIG. 5. The voltage test may be performed during manufacture at a variety of pressures and temperatures. The voltage test may be used to ensure an electrochemical portion (e.g., the internal subassembly **100**) of the electrolyzer cell **195** and/or the electrolyzer stack **196** is isolated from the remaining hardware, such as the compression hardware and/or the non-repeating hardware (e.g., the plurality of bolts **177** and/or the end plates **168**, **169**). Unwanted communication between the electrochemical portion (e.g., the internal subassembly **100**) and the remaining and/or non-repeating hardware may present a hazard during

operation, such as electrical shorting of the electrolyzer cell **195** and/or the electrolyzer stack **196**. Such unwanted communication between the electrochemical portion and the remaining and/or non-repeating hardware may also present a risk of electrocution for persons such as operators of the system **1** who may be nearby the electrolyzer cell **195** and/or the electrolyzer stack **196**. For the voltage test, a high-voltage power supply may be used to apply a potential to the electrolyzer cell **195** and/or the electrolyzer stack **196**.

[0054] Where a voltage test is performed, the system **1** may include a voltage-testing device **350**. In some embodiments, the voltage-testing device **350** may include a sensor, such as an amperage meter, to monitor the current being supplied by the power supply. In embodiments, the voltage testing device **350** may further include a controller, transmitter and/or receiver (not shown). The controller may be used to regulate the current and/or voltage applied to the electrolyzer cell **195** and/or the electrolyzer stack **196**. The transmitter may transmit an indication of the cell voltage to the controller. The receiver may receive a signal directing the voltage-testing device to increase, decrease, maintain or cease the voltage applied to the electrolyzer cell **195** and/or the electrolyzer stack **196**. The voltage-testing device **350** may be removably connectable to various layers of the electrolyzer cell **195** and/or the electrolyzer stack **196**. For example, the voltage-testing device **350** may be removably connected to an interior, electrochemical portion (e.g., subassembly **100**) of the electrolyzer cell **195** and/or the electrolyzer stack **196** in addition to exterior hardware such as any of the plurality of bolts **177** (e.g., the first bolt **177a**). The voltage-testing device **350** may monitor the electrical current traveling between and/or through the various layers to assist in determining whether there is unwanted electrical and/or electrochemical communication between any of the various components of the electrolyzer cell **195** and/or the electrolyzer stack **196**. Pass/fail criteria for the voltage test may be based on the maximum current amplitude, rate of change or other measurable change in the electrical stimuli.

[0055] Advantageously, the leak-check, the voltage test, and various other tests which one skilled in the art would appreciate as being useful in determining the operational status of the electrolyzer stack or a portion thereof (e.g., the electrolyzer cell **195**) may be performed while the electrolyzer cell **195** and/or the electrolyzer stack **196** (e.g., the two separators **170**, **171**, the two structural frames **172**, **173**, the sealant **190**, and/or the other components of the internal subassembly **100**) is still connected to the system **1**, such as to the tensioning device **200** and/or the fluid pump **250**. Thus, if the electrolyzer cell **195** and/or the electrolyzer stack **196** fails any test(s), such that it is determined that the electrolyzer cell **195** and/or the electrolyzer stack **196** requires further processing, the electrolyzer cell **195** and/or the electrolyzer stack **196** may still be connected to the manufacturing components of the system **1** and may continue to be processed without undue and/or redundant efforts on the part of the operator(s).

[0056] With reference to FIG. 7, a method of the present disclosure will now be discussed. Specifically, the method pertains to operating the system **1** described above. Said another way, disclosed herein is a method for manufacturing and/or sealing an electrolyzer cell (e.g., the electrolyzer cell **195**) of an electrolyzer stack (e.g., the electrolyzer stack **196**).

[0057] In embodiments, the method of the present disclosure may include applying the sealant **190** between the two separators **170**, **171** and/or the two structural frames **172**, **173**, as in Step **1** (S1). The sealant **190** may be applied with the intention of conforming the sealant **190** to the two separators **170**, **171** and/or the two structural frames **172**, **173** in order to bind them together.

[0058] In embodiments, the method may further include compressing the two separators **170**, **171** and/or the two structural frames **172**, **173** toward each other, as in Step **2** (S2). The two separators **170**, **171** and/or the two structural frames **172**, **173** may be compressed towards each other and/or towards the sealant **190** to facilitate the conforming of the sealant to the two separators **170**, **171** and/or the two structural frames **172**, **173** as described above. The tension, pressure and/or compression may be applied to the two separators **170**, **171**, the two structural frames **172**, **173** and the sealant **190** by way of the tensioning device **200**, as described above. In the example, the

tensioning device **200** may be a hydraulic bolt tensioner, which may be used to tighten one or multiple bolts of the plurality of bolts **177**. In some embodiments, the tensioning device **200** may only be able to apply compression to the first bolt **177a** and not to the remaining bolts of the plurality of bolts **177**. Under such a circumstance, it may be prudent to apply compression using multiple of the tensioning device **200**, each of which acting on one bolt of the plurality of bolts **177**. In other embodiments, the pressure applied by one of the tensioning device **200** to one or multiple of the plurality of bolts **177** may be sufficient to carry out the method of operating the system **1**. In other embodiments, the force or compression may be applied independent of any of the plurality of bolts **177** (such as putting the electrolyzer cell **195** and/or the electrolyzer stack **196** in a hydraulic press). The compressing the two separators **170**, **171** and/or the two structural frames **172**, **173** towards each other may include setting parameters for active pressure regulation and/or control using the tensioning device **200** (e.g., the pressure controller **202**) during manufacturing. [0059] In embodiments, the method may further include connecting the fluid pump **250** to the fluid port **180** to permit the fluid pump **250** to flow fluid through the one or multiple channels and/or the flow field(s) **175**, **176** of the two structural frames **172**, **173**, as in Step 3 (S3).

[0060] In embodiments, the method may further include flowing the fluid through the one or multiple channels of the two structural frames **172**, **173**, as in Step 4 (S4). The flowing of the fluid through the one or multiple channels and/or the flow field(s) **175**, **176** may include setting parameters for active regulation and/or control of flow and/or temperature control using the pump controller **252** during manufacturing. In some embodiments, the flowing fluid through the one or multiple channels and/or the flow field(s) **175**, **176** may further include gradually increasing the temperature of the fluid to reach a process temperature. The process temperature may be the temperature at which the sealant **190** is best able to conform to the two separators **170**, **171** and/or the two structural frames **172**, **173** as desired. The process may maintain prescribed temperatures throughout operation and may be able to maintain the desired characteristics within acceptable ranges. This may include actively increasing or decreasing temperatures to best suit the product and/or process. This temperature regulation can be done via heater, chillers, multi-fluid heat exchangers etc.

[0061] In embodiments, the method may further include actively controlling the pressure applied to the two separators **170**, **171**, the two structural frames **172**, **173**, and the sealant **190** during operation of the system **1**, as in Step 5 (S5). In some embodiments, the controlling the pressure may be performed manually, such as by an operator who is fed information from the pressure sensor **204**. The operator may regulate and/or control the pressure applied by the system **1** by using the information from the sensor **204** to instruct the pressure controller **202** to modify or maintain the pressure within the proscribed range. Alternatively, the controlling the pressure may be performed automatically and/or dynamically by the pressure controller **202** modifying or maintaining pressure within the proscribed range, as described above with respect to the system **1**.

[0062] In embodiments, the method may further include actively controlling the temperature and/or flow of fluid through the one or multiple channels and/or the flow field(s) **175**, **176**, as in Step 6 (S6). In some embodiments, the controlling the temperature and/or flow may be performed manually, such as by an operator who is fed information from the flow sensor **254** and/or the temperature sensor **256**. The operator may regulate and/or control the flow rate and/or the temperature of the fluid flowed through the one or multiple channels and/or the flow field(s) **175**, **176** by using information from the flow sensor **254** and/or the temperature sensor **256** to instruct the pump controller **252** to modify or maintain the pressure within the proscribed range(s). Alternatively, the controlling the flow rate and/or temperature may be performed automatically or dynamically by the pump controller **252** modifying or maintaining the flow rate and/or temperature of the fluid to be within the proscribed range(s), as described above with respect to the system **1**. In some embodiments, the method may include both S5 & S6 to ensure and/or achieve the desired result.

[0063] In embodiments, the method may further include conforming the sealant **190** to the two separators **170**, **171** and/or the two structural frames **172**, **173**, as in Step **7** (S7). The conforming the sealant **190** may take place during various other steps of the method. For example, the sealant **190** may be conformed, among other ways, in response to the initial pressure applied to the two separators **170**, **171**, the two structural frames **172**, **173**, and the sealant **190** during operation of the system **1**. The sealant **190** may also be conformed in response to the heating of the system by the pump controller **250** flowing fluid through the one or multiple channels and/or the flow field(s) **175**, **176**. At various points during operation of the system **1**, the sealant **190** may conform at different rates. In embodiments, the system **1** may be cooled down following the conforming of the sealant **190** to the separators **170**, **171** and/or the two structural frames **172**, **173**. Said another way, the method may include cooling the system **1** following the conforming the sealant **190**.

[0064] In embodiments, the method may further include performing a leak-check, as in Step **8** (S8). As noted above, the method of the present disclosure advantageously allows for the leak-check to be performed during manufacturing and before the electrolyzer cell **195** and/or the electrolyzer stack **196** is separated or removed from the system **1**. The leak-check may also be performed at various pressures, including operational pressures. The performing the leak-check may include forcing fluid (e.g., water) or gas through the one or multiple channels and/or the flow field(s) **175**, **176** and determining whether unwanted fluid communication is present between any of the channels of the one or multiple channels (e.g., any portions of the flow field(s) **175**, **176**), or between any of the channels (e.g., any portion of the flow field(s) **175**, **176**) and a space exterior to the electrolyzer cell **195** and/or the electrolyzer stack **196**. In embodiments, the performing the leak check may include flowing the fluid or gas through the one or multiple channels and/or the flow field(s) **175**, **176** using the fluid pump **250**. In other embodiments, performing the leak check may include flowing the fluid or gas through the one or multiple channels and/or the flow field(s) **175**, **176** using a different leak-checking device that one of ordinary skill in the art would appreciate as being able to detect leaking fluid or gas. The performing the leak check may also include flowing the fluid or gas through the one or multiple channels and/or the flow field(s) **175**, **176** to check for leaking fluid or gas at various flow rates.

[0065] In embodiments, the method may further include performing a voltage test, as in Step **9** (S9). As noted above, the method of the present disclosure advantageously allows for the voltage test to be performed during manufacturing and before the electrolyzer cell **195** and/or the electrolyzer stack **196** is separated and/or removed from the system **1**. The voltage test may also be performed at various pressures, including operational pressures. The performing the voltage test may include connecting the voltage-testing device **350** to any of the various layers of the electrolyzer cell **195** (e.g., layers of the subassembly **100** (FIG. **1**)) and/or the electrolyzer stack **196**. For example, the performing the voltage test may include connecting the voltage-testing device **350** to an interior, electrochemical portion of the electrolyzer stack **195** in addition to exterior hardware such as any of the plurality of bolts **177** (e.g., the first bolt **177a**). The performing the voltage test may further include monitoring the electrical current traveling between and/or through the various layers to determine whether there is unwanted electrical and/or electrochemical communication between any of the various components of the electrolyzer cell **195** and/or the electrolyzer stack **196**.

[0066] In some embodiments, should the electrolyzer cell **195** and/or the electrolyzer stack **196** fail any test performed thereon (e.g., the leak-check and/or the voltage test, among other tests), the method may further include repeating Steps **4-9**. Steps **4-9** may be repeated until the electrolyzer cell **195** and/or the electrolyzer stack **196** is deemed and/or determined to be in operating condition. The method may thus include determining whether the electrolyzer stack **195** is in operational condition for use in electrolysis. In other embodiments repair of various components of the electrolyzer cell **195** and/or the electrolyzer stack **196** following a testing failure may require rework prior to the listed S1 step, where the entire sequence would have to be repeated.

[0067] In embodiments, the method may further include releasing the compression applied by the tensioning device **200** and removing the completed electrolyzer stack **195**, as in Step **10** (**S10**). In some embodiments, the releasing the compression and removing the completed electrolyzer stack **195** may only occur following a determination that the electrolyzer stack **195** is in operational condition for use in electrolysis.

[0068] In embodiments, the method may further include preparing the system **1** to perform the method again using a second set of two separators, a second set of structural frames, and a second sealant (e.g., another of the two separators **170**, **171**, the two structural frames **172**, **173**, and the sealant **190**). Once the finished electrolyzer cell **195** and/or electrolyzer stack **196** has been removed from the system **1** and before the subsequent stack components have been placed into the system **1**, the system **1** may require a servicing and/or cleaning procedure to be performed. In other embodiments no process may be required to prepare the system **1** to seal the next electrolyzer cell **195** and/or electrolyzer stack **196**. Regardless, it is imperative that the system **1** be free of contaminants to avoid poisoning subsequently manufactured electrolyzer cells and/or stacks.

[0069] As may be recognized by those of ordinary skill in the art based on the teachings herein, numerous changes and modifications may be made to the above-described and other embodiments of the present disclosure without departing from the scope of the disclosure. The components of the system and method as disclosed in the specification, including the accompanying abstract and drawings, may be replaced by alternative component(s) or feature(s), such as those disclosed in another embodiment, which serve the same, equivalent or similar purpose as known by those skilled in the art to achieve the same, equivalent or similar results by such alternative component(s) or feature(s) to provide a similar function for the intended purpose. In addition, systems and methods described herein may include more or fewer components or features than the embodiments as described and illustrated herein. For example, the components and features of FIGS. 3-6 may be used interchangeably and in alternative combinations as would be modified or altered by one of ordinary skill in the art. Further, the systems and methods disclosed herein may also be useful in producing fuel cells in addition to electrolyzer cells as described above. Accordingly, this detailed description of the currently preferred embodiments is to be taken illustratively, as opposed to limiting the disclosure.

[0070] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprise” (and any form of comprise, such as “comprises” and “comprising”), “have” (and any form of have, such as “has”, and “having”), “include” (and any form of include, such as “includes” and “including”), and “contain” (and any form of contain, such as “contains” and “containing”) are open-ended linking verbs. As a result, a method or system that “comprises,” “has,” “includes,” or “contains” one or more steps or elements possesses those one or more steps or elements but is not limited to possessing only those one or more steps or elements. Likewise, a step of a method or an element of a device that “comprises,” “has,” “includes,” or “contains” one or more features possesses those one or more features but is not limited to possessing only those one or more features. Furthermore, a device or structure that is configured in a certain way is configured in at least that way but may also be configured in ways that are not listed.

[0071] The disclosure has been described with reference to the preferred embodiments. It will be understood that the architectural and operational embodiments described herein are exemplary of a plurality of possible arrangements to provide the same general features, characteristics, and general system operation. Modifications and alterations will occur to others upon a reading and understanding of the preceding detailed description. It is intended that the disclosure be construed as including all such modifications and alterations.

Claims

1. A method for sealing an electrolyzer cell, the method comprising: applying a sealant between two layers of an electrolyzer cell; compressing the two layers towards each other; flowing fluid through a flow field in the electrolyzer cell; controlling a temperature of the fluid flowing through the flow field; controlling a pressure applied to the sealant by the compressing the two layers towards each other; and conforming the sealant to the two layers by the controlling the temperature and the controlling the pressure.
2. The method of claim 1, wherein the controlling the pressure further comprises automatically controlling the pressure of the compressing the two layers towards each other such that the pressure is maintained within a prescribed range during the conforming the sealant to the two layers.
3. The method of claim 1, wherein the controlling the pressure further comprises automatically controlling the pressure of the compressing the two layers towards each other to control the conforming the sealant to the two layers.
4. The method of claim 1, wherein the compressing the two layers towards each other further comprises compressing the two layers towards each other using a tensioning device, and wherein the method further comprises releasing the tensioning device and removing the two layers from the tensioning device.
5. The method of claim 1, wherein the controlling the temperature of the fluid further comprises gradually increasing the temperature of the fluid to reach a threshold temperature using a heater.
6. The method of claim 4, further comprising the tensioning device receiving a second set of two layers.
7. The method of claim 1, wherein the electrolyzer cell is an electrolyzer cell of an electrolyzer stack, and wherein the method further comprises: connecting a voltage tester to an electro-chemical portion of the electrolyzer cell; connecting the voltage tester to a portion of a remainder of the electrolyzer stack, the remainder including any portion of the electrolyzer stack other than the electro-chemical portion; and measuring an electrical current between the electro-chemical portion and the portion of the remainder.
8. The method of claim 1, further comprising checking for leaking fluid between two portions of the flow field after the compressing the two layers towards each other.
9. The method of claim 8, wherein the checking for leaking fluid further comprises checking for leaking fluid at an operational pressure, the operational pressure being the pressure at which the two layers are intended to function in the electrolyzer cell during electrolysis.
10. The method of claim 1, further comprising checking for leaking fluid or gas between a portion of the flow field and an exterior space surrounding the two layers.
11. The method of claim 10, wherein the checking for leaking fluid or gas further comprises checking for leaking fluid or gas at an operational pressure, the operational pressure being the pressure at which the two layers are intended to function in the electrolyzer cell during electrolysis.
12. A system for sealing an electrolyzer cell, the system comprising: a fluid pump releasably connectable to a flow field of an electrolyzer cell, the electrolyzer cell including two layers having a sealant therebetween, wherein the fluid pump is configured to flow fluid through the flow field; a thermal controller configured to control a temperature of the fluid flowed through the flow field; and a tensioning device releasably connectable to the electrolyzer cell, the tensioning device configured to compress the two layers towards each other while the fluid flows through the flow field and the temperature of the fluid is controlled to conform the sealant between the two layers.
13. The system of claim 12, wherein the tensioning device is further configured to automatically control the pressure applied to the two layers to be within a prescribed range.
14. The system of claim 12, wherein the tensioning device is further configured to automatically control the pressure applied to the two layers to control a conforming of the sealant to the two

layers.

15. The system of claim 12, wherein the thermal controller is further configured to gradually increase the temperature of the fluid flowed through the flow field to a threshold temperature.

16. The system of claim 12, wherein the electrolyzer cell is an electrolyzer cell of an electrolyzer stack, and wherein the system further comprises: a voltage tester releasably connectable to an electro-chemical portion of the electrolyzer cell and to a portion of a remainder of the electrolyzer stack, the remainder of the electrolyzer stack including any portion of the electrolyzer stack other than the electro-chemical portion; and wherein the voltage tester is configured to measure an electrical current between the electro-chemical portion and the portion of the remainder.

17. The system of claim 12, further comprising a leak-checking device releasably connectable to the flow field and configured to check for fluid leaking between two portions of the flow field.

18. The system of claim 17, wherein the leak-checking device is further configured to check for fluid leaking between the two portions of the flow field at an operational pressure, the operational pressure being at least as much pressure as the two layers are intended to function under within the electrolyzer cell during electrolysis.

19. The system of claim 12, further comprising a leak-checking device releasably connectable to the flow field and configured to check for fluid leaking between any portion of the flow field and an exterior space surrounding the two layers.

20. The system of claim 19, wherein the leak-checking device is further configured to check for fluid leaking between the any portion of the flow field and the exterior space at an operational pressure, the operational pressure being at least as much pressure as the two layers are intended to function under within the electrolyzer cell during electrolysis.
